



■ PRESENTED BY
Shafira Husna Rofananda

SPORTS FOOTWEAR SALES & CONSUMER BEHAVIOR ANALYSIS

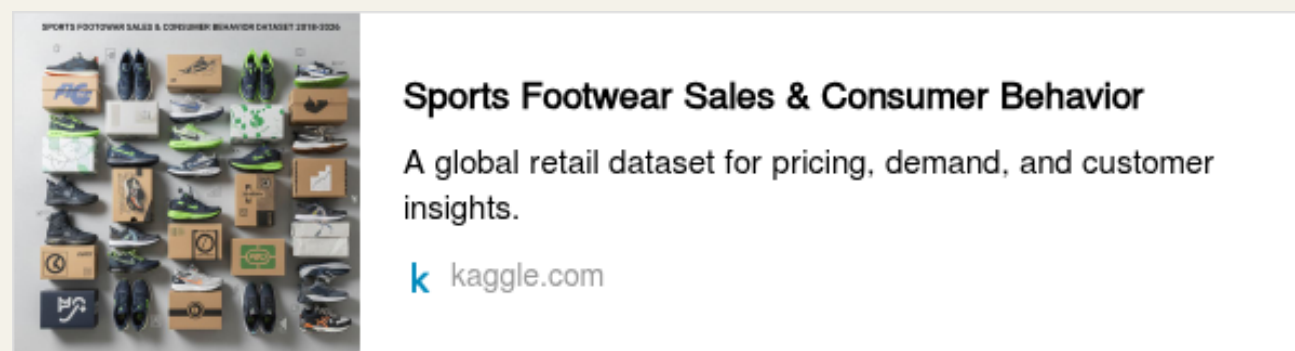
2025 Sales & Consumer Behavior Analysis





INTRODUCTION

The **Footwear.csv** dataset contains **sales, pricing, and customer data** for athletic footwear from **2018 – 2026**. The dataset includes several global brands, including Nike, Adidas, Puma, Reebok, ASICS, and New Balance. The dataset consists of **18 columns** and **30,000 rows**. Based on this dataset, **sales, pricing, and customer behavior analyses** were conducted to support business decision-making.





COLUMN LIST

Column	Explanation
order_id	Uniq ID for each sale
order_date	Sale date
brand	Brand of the shoe
model_name	Spesific model of the shoe
category	Category of the shoe
gender	Men, women, unisex
size	Shoe size
color	Color of the shoe
base_price_usd	Original price

Column	Explanation
discount_percent	Discount applied (%)
final_price_usd	Price after discount
units_sold	Quantity sold
revenue_usd	Total revenue for the order
payment_method	Card, cash, wallet, bank transfer
sales_channel	Online, retail store
country	Country of sale
customer_income_level	Low, medium, high
customer_rating	Rating



PROJECT OBJECTIVE

- Analyze **sales performance**
- Identify **customer purchasing behavior**
- Evaluate **pricing and discount patterns**
- Discover **key business insights** from sales data





DATA CLEANING

Checking and Handling:

- Duplicate Data
- Outliers
- Missing Values



DUPLICATE DATA

```
len(df.drop_duplicates()) / len(df)
```

```
1.0
```

The duplicate checking process shows that the ratio between unique rows and total rows is 1, indicating that **no duplicate data** was found in the dataset.

MISSING VALUES

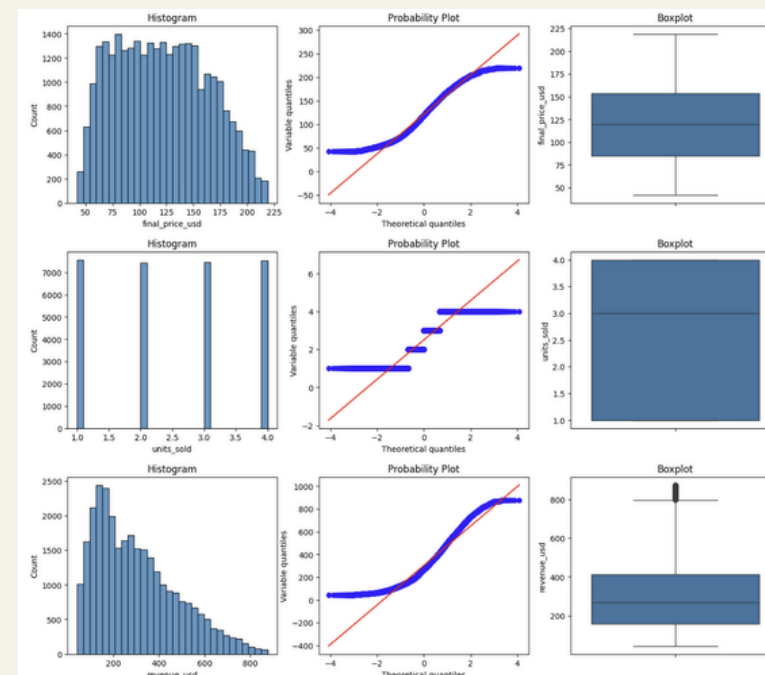
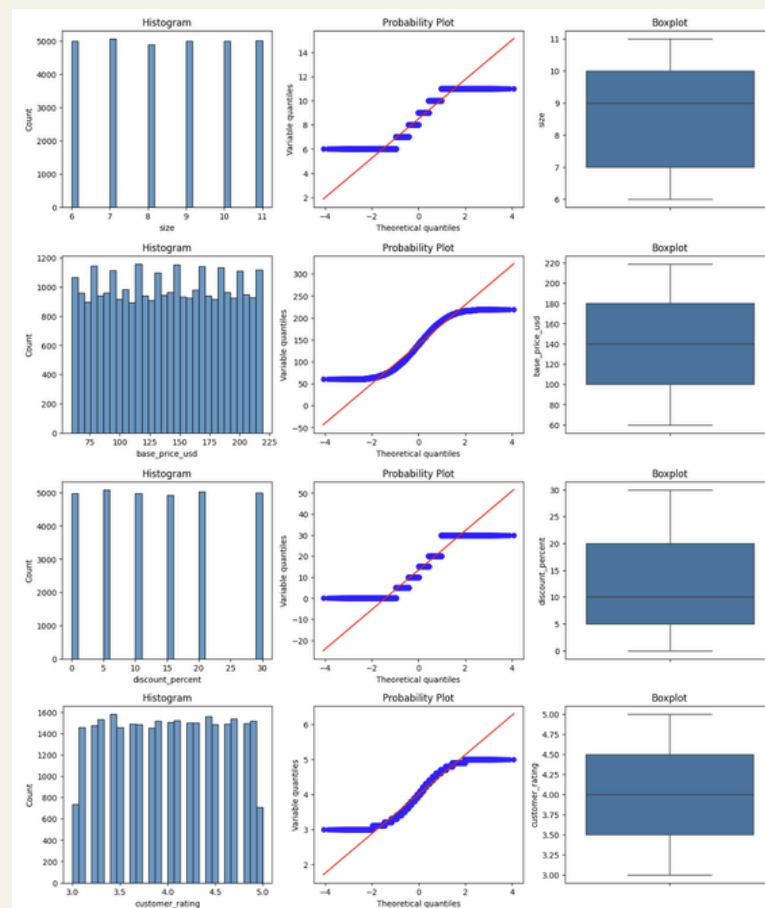
```
df.isna().sum()
```

```
df.isnull().sum()
```

```
for column in df.columns:
    print(f"===== {column} =====")
    display(df[column].value_counts())
    print()
```

The dataset contains **no missing values** and all unique values in each column are consistent and appropriate with the dataset context.

OUTLIERS



Based on the visualization results using histograms, probability plots, and boxplots on numerical columns, all variables **did not show outliers**. Although several relatively high values were found in the revenue_usd variable, these values were still considered reasonable within the business context and therefore were retained and not treated as outliers.

DATA TRANSFORM

1. Convert the order_date column data type to datetime

```
df["order_date"] = pd.to_datetime(df["order_date"], format="%Y-%m-%d")
```

2. Convert the customer_income_level column data type to categorical

```
order = ['Low', 'Medium', 'High']  
df['customer_income_level'] = pd.Categorical(df['customer_income_level'],  
                                             categories=order, ordered=True)
```

3. Create date, month, and year columns

```
df["day"] = df["order_date"].dt.day  
df["month"] = df["order_date"].dt.month  
df["year"] = df["order_date"].dt.year
```


EXPLANATORY DATA ANALYSIS

The **Exploratory Data Analysis** (EDA) process was conducted to:

- Analyze **sales performance**
- Identify **customer purchasing behavior**
- Evaluate **pricing and discount patterns**
- Discover **key business insights** from sales data

Although the dataset contains footwear sales data from 2018 to 2026. **This analysis focuses specifically on sales data from 2025.**





KPI

Order	YoY Order	Total Quantity	YoY Quantity
3351	4.59%	8483	5.47%
Total Revenue	YoY Revenue	AOV	YoY AOV
1.02M	5.57%	305.61	0.94%

Overall KPI performance in 2025 showed positive growth compared to 2024.

ORDER

Total orders reached 3351, representing a year-over-year (YoY) increase of 4.59%.

TOTAL QUANTITY

Total quantity reached 8483, representing a year-over-year (YoY) increase of 5.47%.

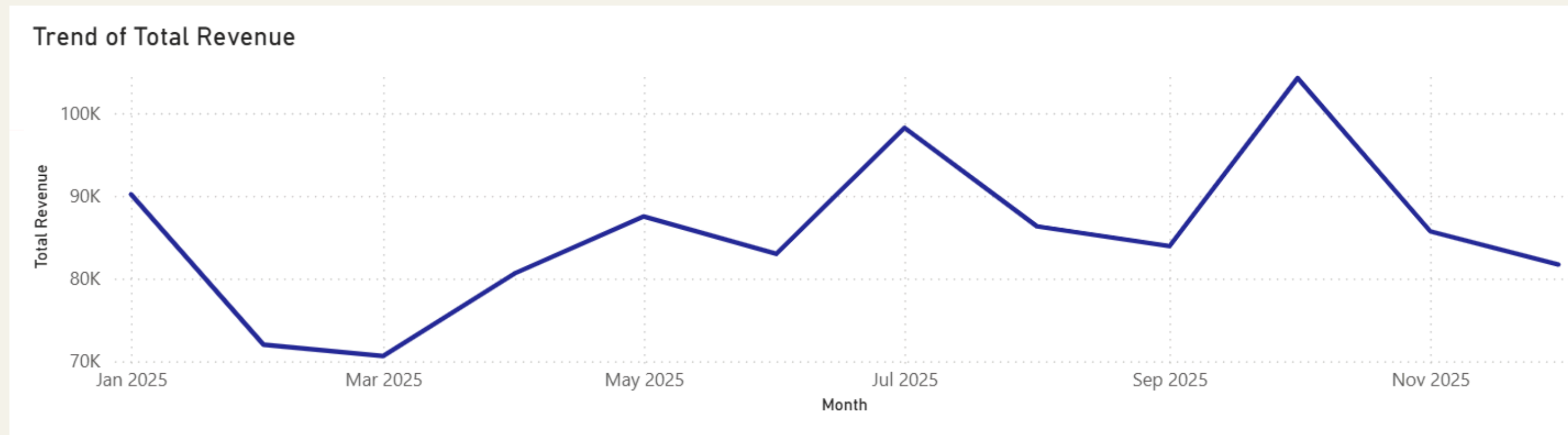
TOTAL REVENUE

Total revenue reached USD 1.02 M, representing a year-over-year (YoY) increase of 5.57%.

AOV

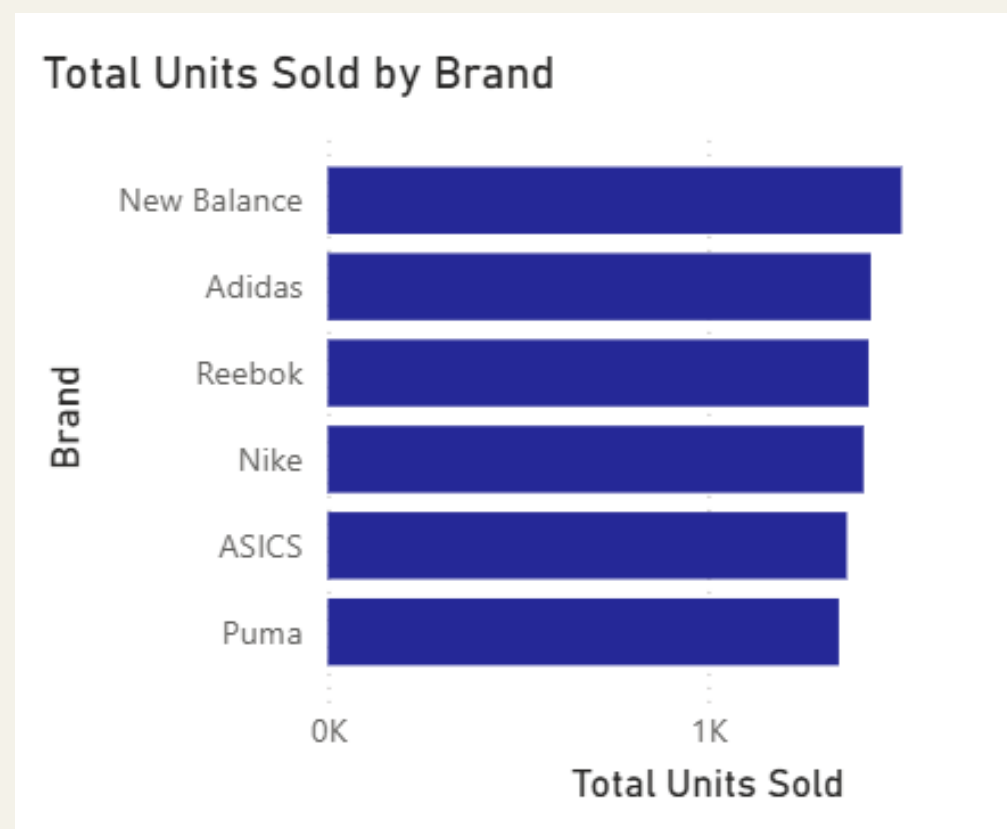
AOV reached USD 305.61, representing a year-over-year (YoY) increase of 0.94%.

Revenue trends throughout 2025 appeared to fluctuate without a clear or consistent pattern.

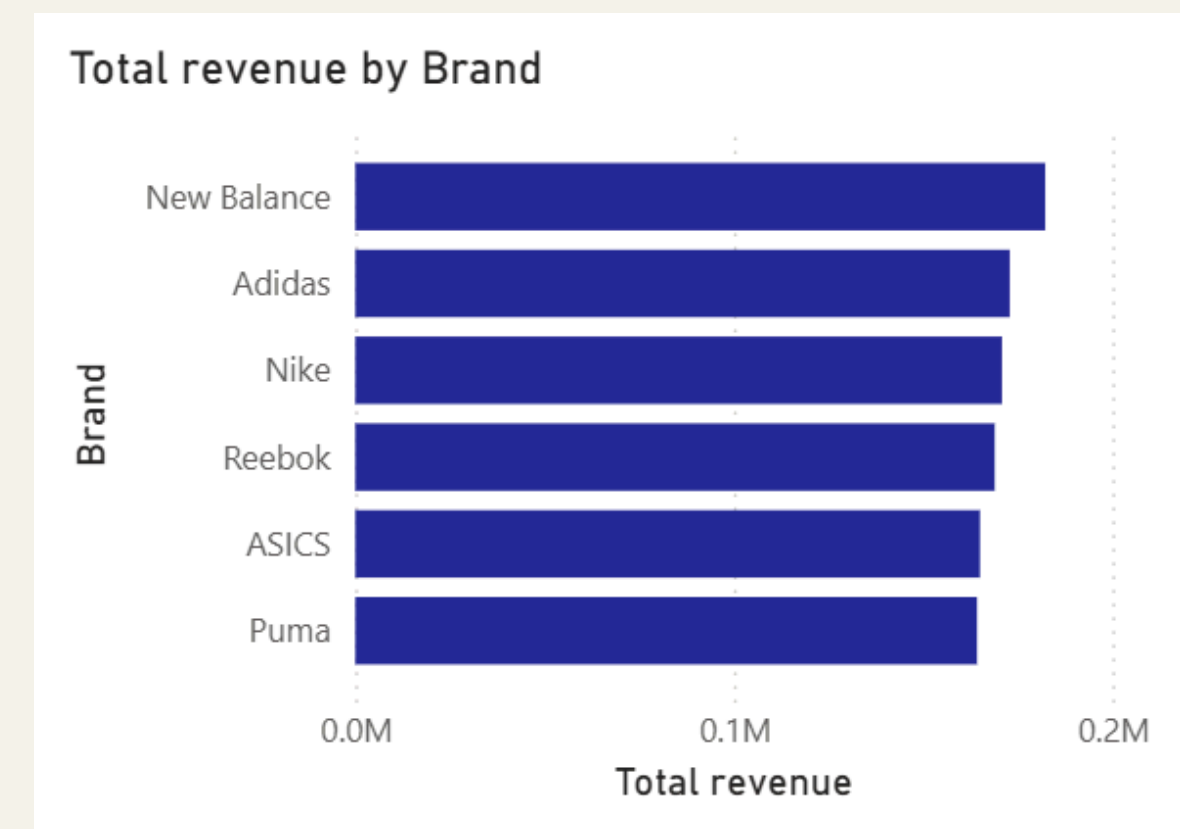


The lowest revenue was recorded in March, while the highest revenue occurred in October. This indicates that sales performance varied across months without showing a steady increase or decrease.

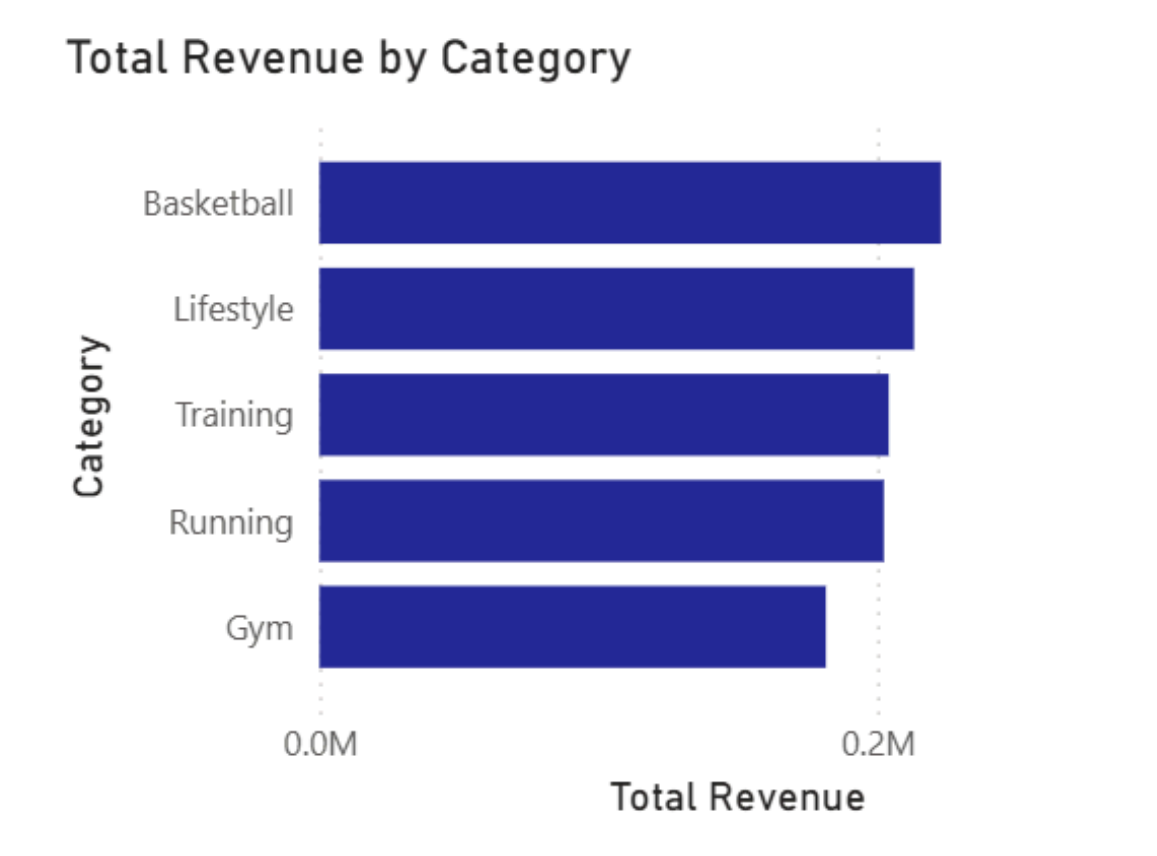
New Balance dominated both revenue and units sold in 2025, indicating strong market demand and sales performance.



New balance recorded the highest sales volume in 2025

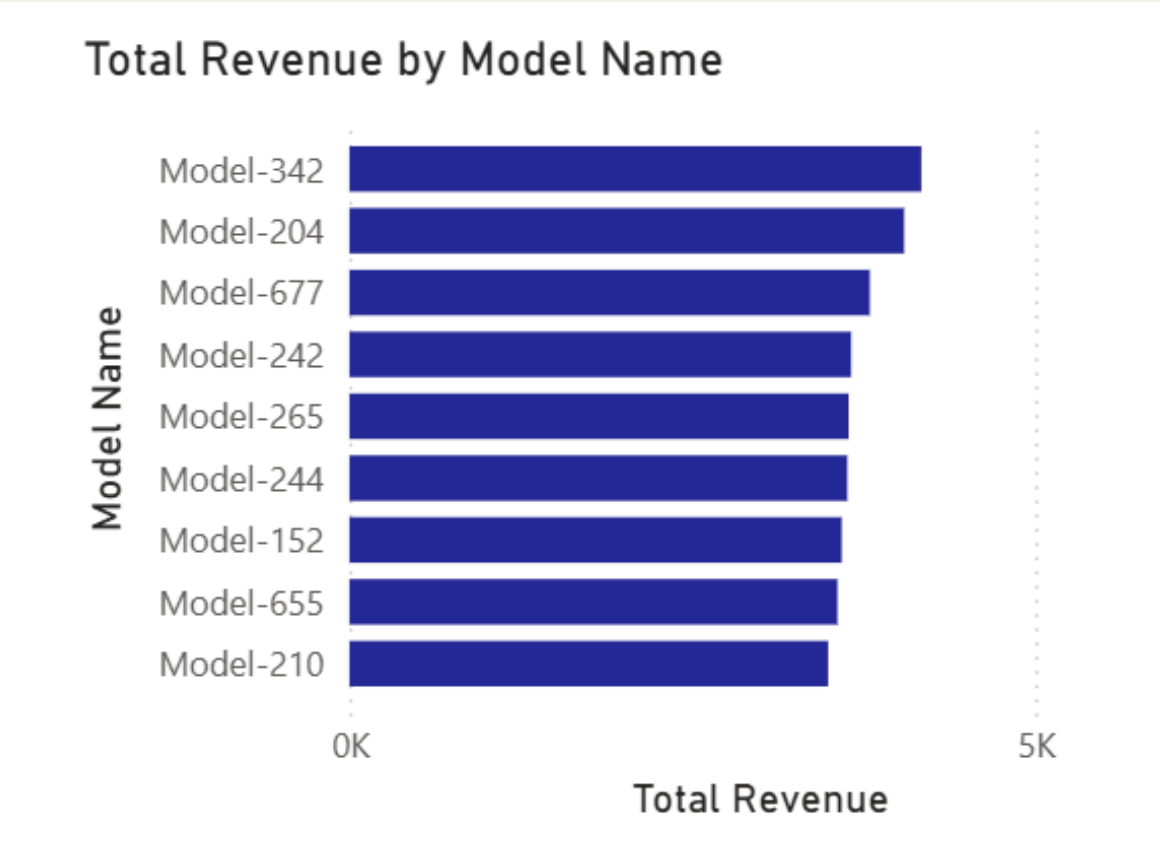


New Balance generated the highest revenue among all brands in 2025



Brand	Top Category
Adidas	Lifesyle
ASICS	Basketball
New Balance	Lifestyle
Nike	Basketball
Puma	Basketball
Reebok	Basketball

Basketball Shoes generated the highest overall revenue in 2025. However, category performance varied across brands. While several brands were dominated by Basketball Shoes, other brands generated higher revenue from Lifestyle Shoes, indicating differences in customer preferences and brand positioning.



Brand	Top Model
Adidas	Model-820
ASICS	Model-760
New Balance	Model-152
Nike	Model-491
Puma	Model-242
Reebok	Model-427

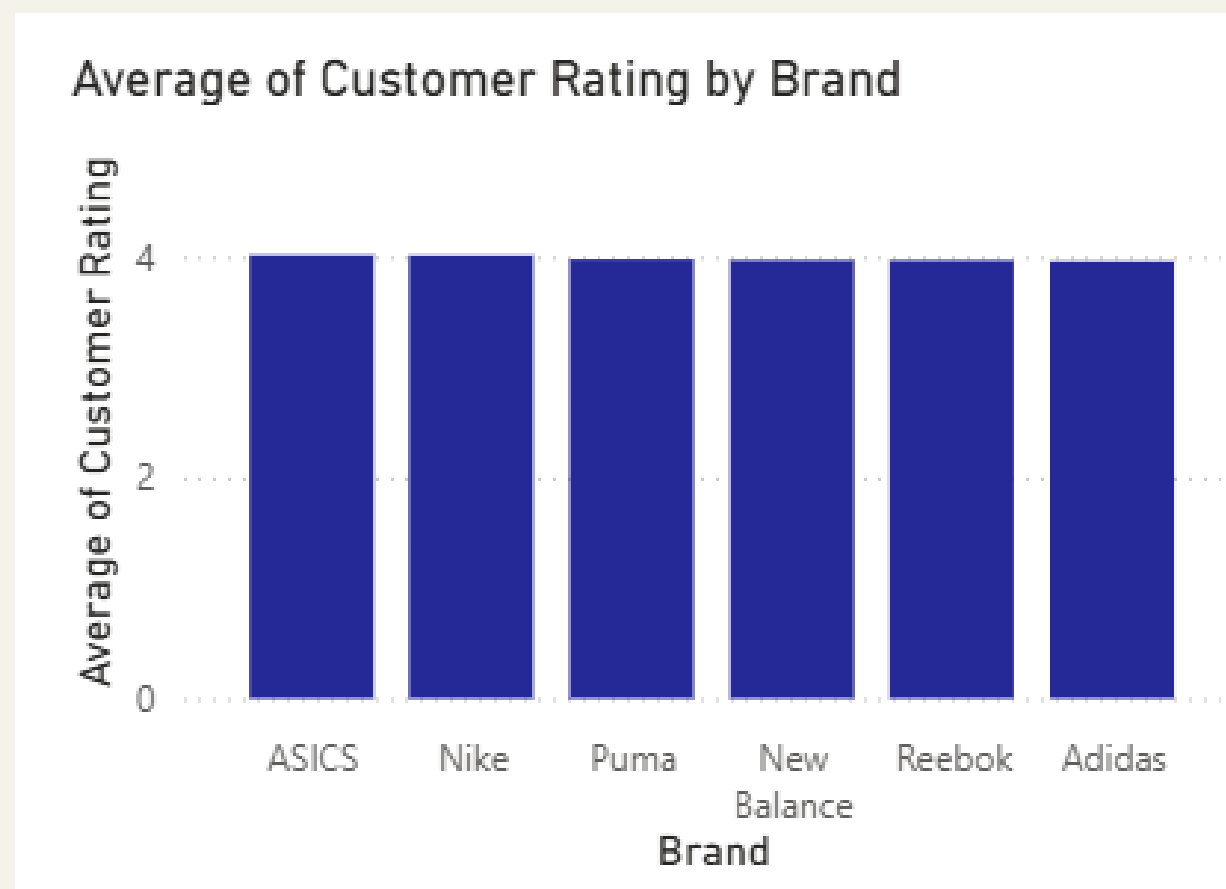
Model-342 generated the highest overall revenue in 2025. However, model performance varied across brands with each brand showing different top-performing models, indicating that brands have their own flagship products and unique customer preferences.

	gender	units_sold
1	Unisex	2936
0	Men	2844
2	Women	2703

Unisex shoes recorded the highest sales volume in 2025.

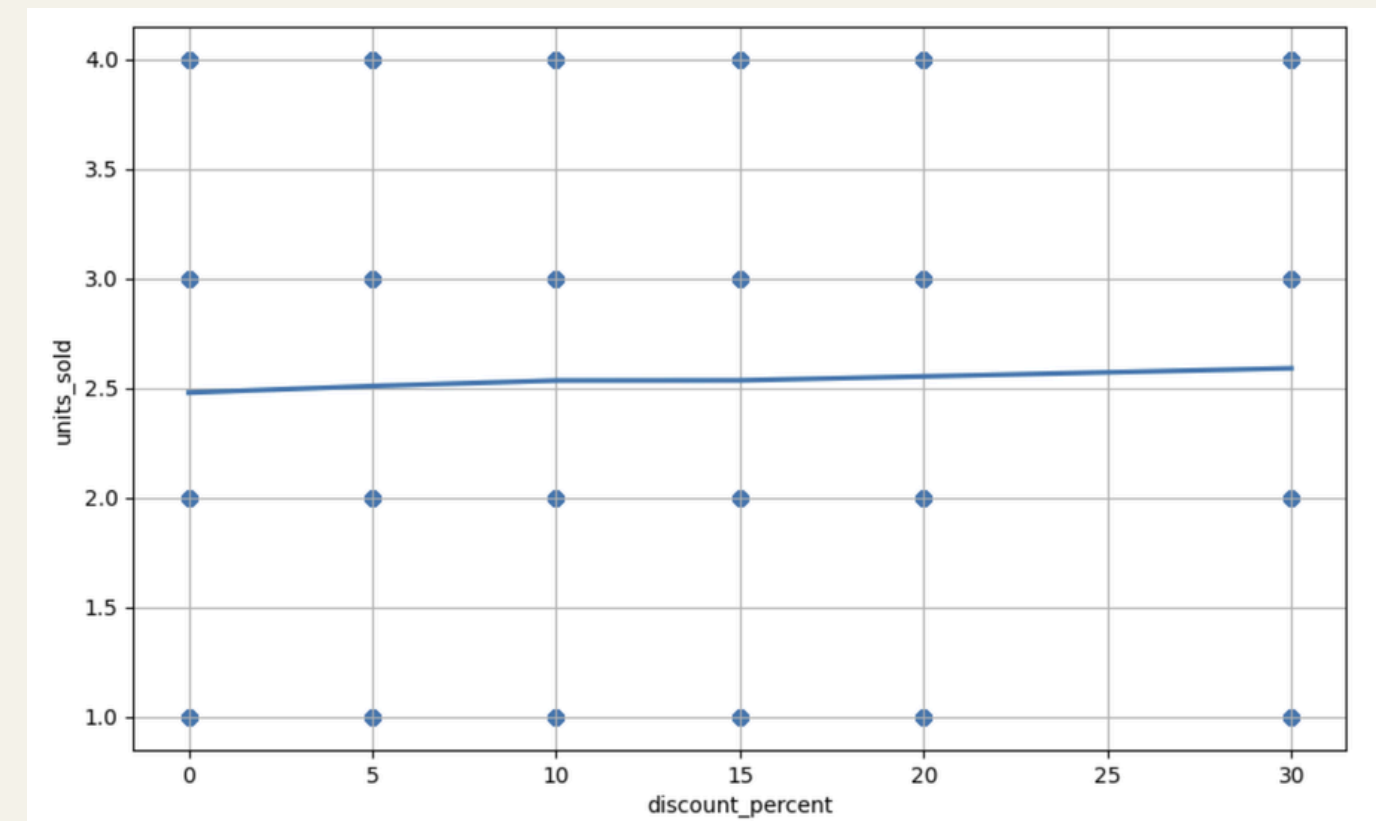
This indicates that customers tend to prefer gender-neutral footwear that can appeal to a broader market without emphasizing specific gender characteristics. This trend is also supported by men's shoes ranking second in sales volume, as many men's footwear designs also feature gender-neutral styles.

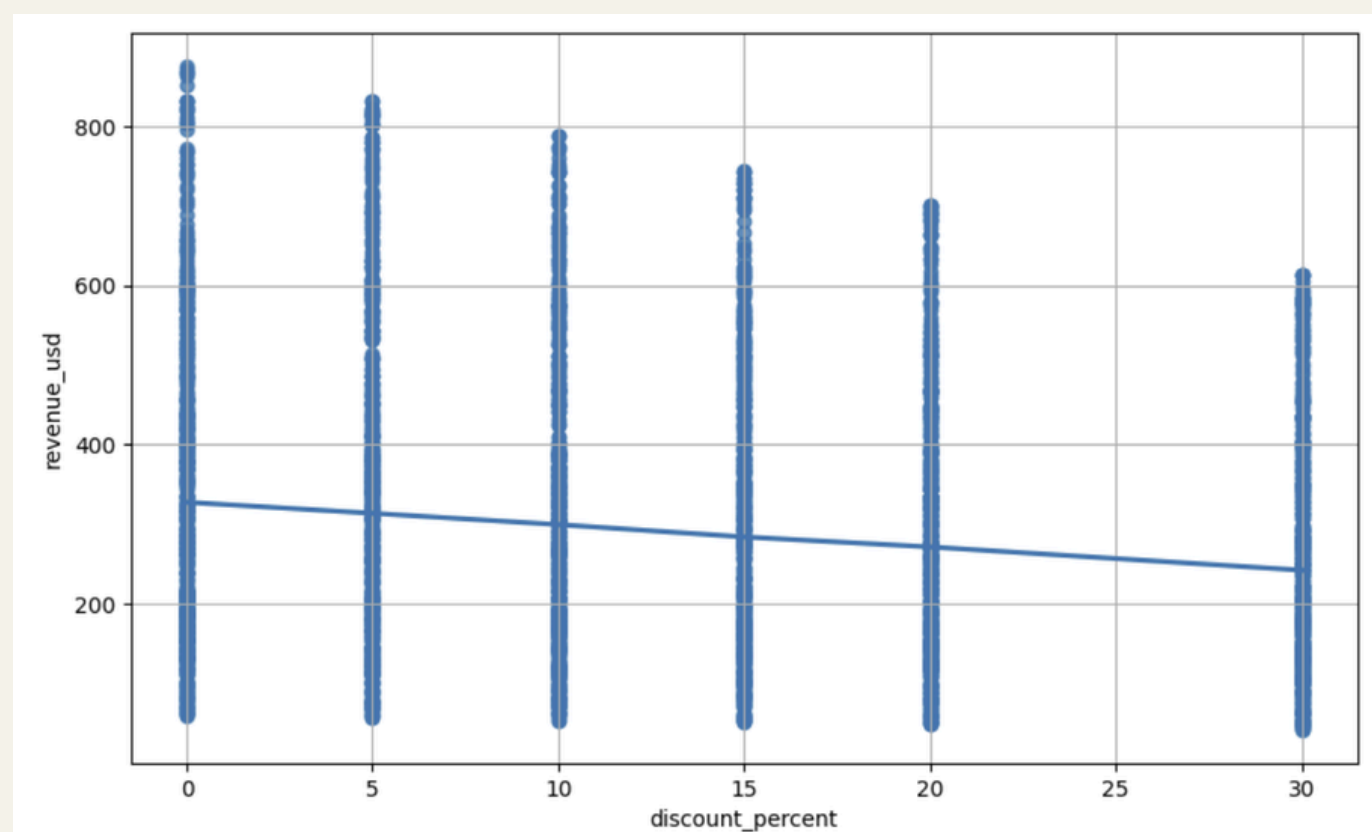
All brands showed a similar average rating in 2025, with scores around 4 out of 5. This indicates that the brands generally maintain good product quality and customer satisfaction levels, with no significant differences in perceived quality among brands.



	brand	customer_rating
0	ASICS	4.015543
3	Nike	4.013251
4	Puma	3.978860
2	New Balance	3.970377
5	Reebok	3.969176
1	Adidas	3.960708

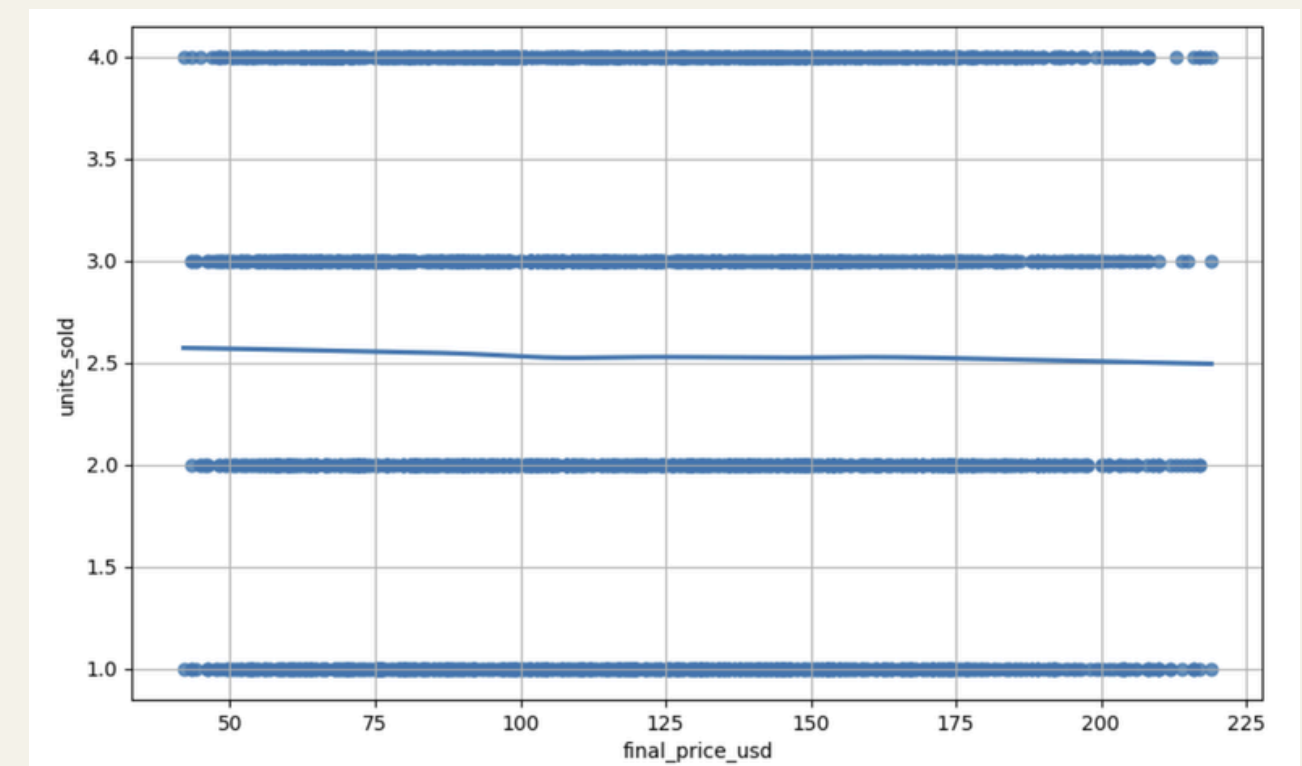
The relationship between discount percentage and units sold in 2025 was positive. Higher discount percentages tended to be associated with an increase in the number of units purchased per transaction. This indicates that larger discounts may encourage customers to buy more footwear products in a single transaction



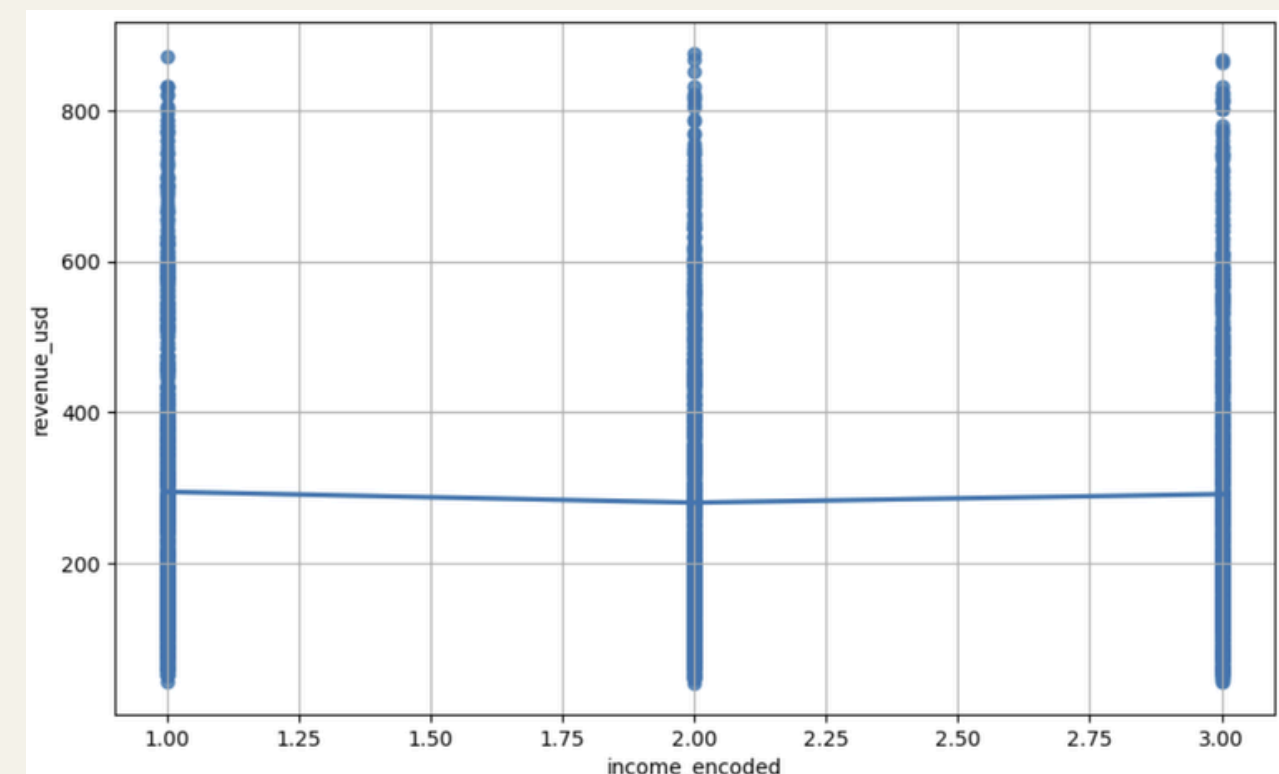
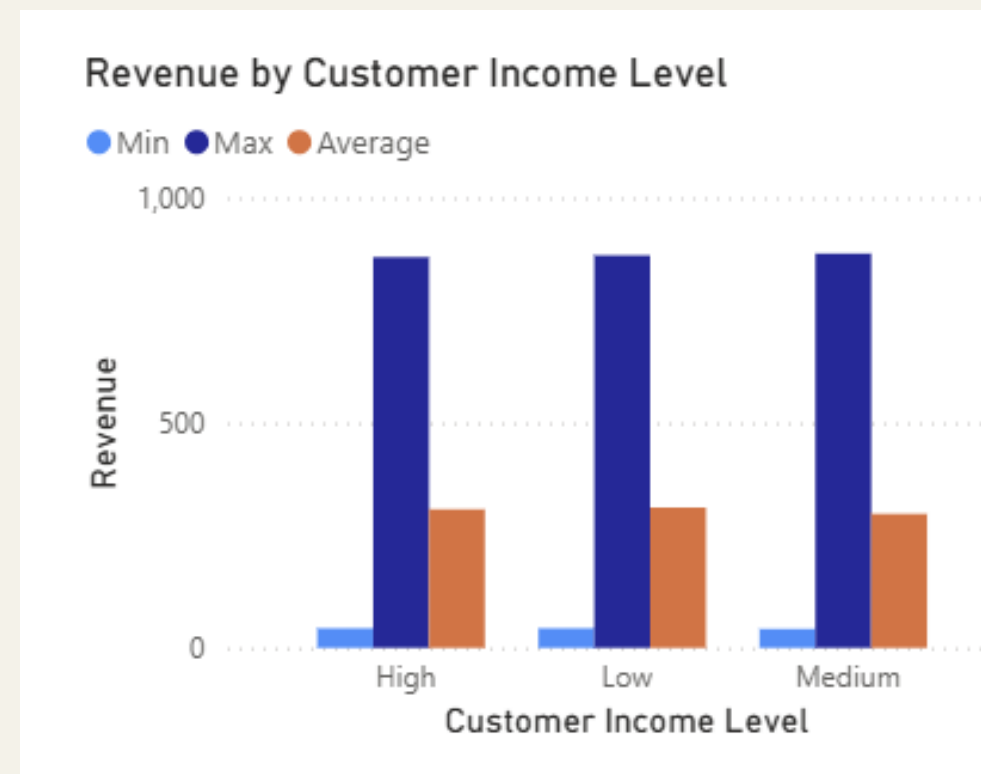


The relationship between discount percentage and revenue in 2025 was negative. Higher discount percentages tended to be associated with lower revenue per transaction. This indicates that although discounts may encourage purchases, larger discounts can reduce the revenue generated from each sale.

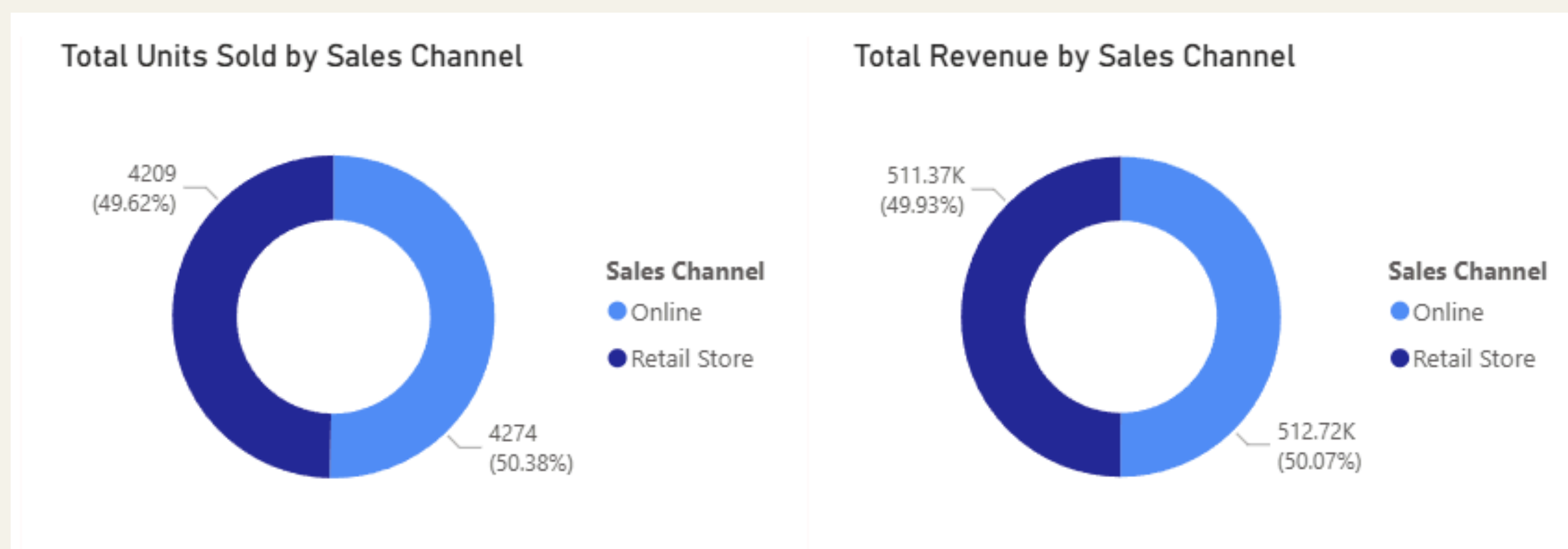
The relationship between final price and revenue in 2025 was negative. Higher final prices tended to be associated with lower revenue per transaction. This may indicate that when prices increase, customers are less likely to purchase footwear products in large quantities within a single transaction.



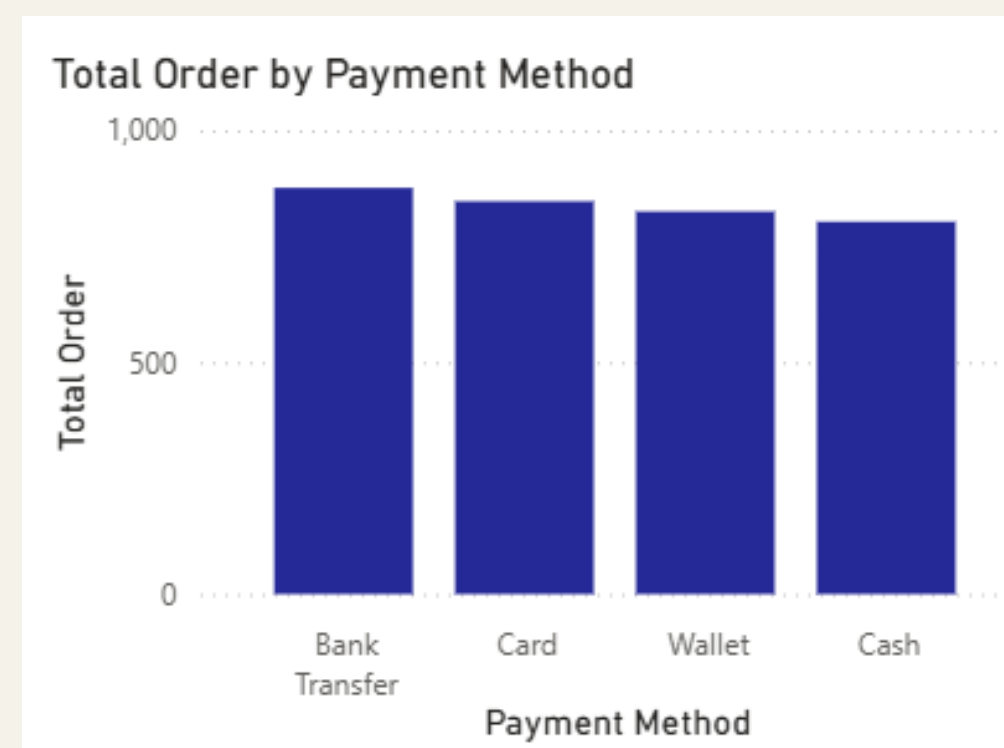
The minimum, maximum, and average revenue generated per transaction were relatively similar across low, medium, and high income customer groups in 2025. This indicates that customer income level may not significantly influence the revenue generated from each transaction



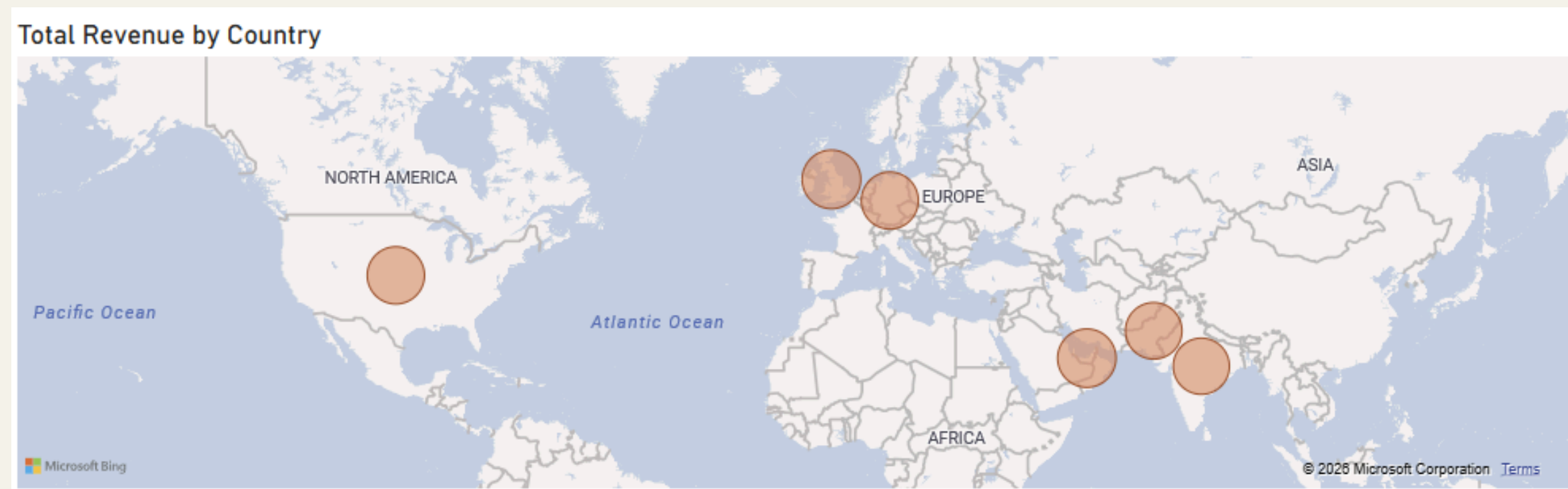
Online sales channels recorded higher performance in both units sold and revenue compared to retail store. However, the difference between online and retail store channels was relatively small, indicating that both channels play equally important roles in overall sales performance.



Bank transfer was the most frequently used payment method in 2025. Compared to card, digital wallet, and cash payments, customers showed a stronger preference for bank transfers, possibly due to convenience, accessibility,



Revenue generated across countries was relatively similar in 2025. This indicates that sales performance and customer demand tended to be relatively consistent across different countries during the analyzed period.



CONCLUSION

- 01 **Business performance improved in 2025** compared to 2024, although monthly revenue trends fluctuated throughout the year.
 - 02 **New Balance** recorded the highest revenue and units sold, indicating strong sales performance and market demand.
 - 03 **Basketball Shoes** generated the highest overall revenue, while each brand showed different top-performing models and categories.
 - 04 **Unisex footwear** recorded the highest sales volume, suggesting strong customer preference for versatile footwear products.
 - 05 **Discount strategies** increased units sold but tended to reduce revenue per transaction.
 - 06 Both **online and retail store channels** contributed significantly to overall sales performance.
-



RECOMMENDATION

- 01 **Optimize promotional strategies** to balance sales growth and revenue performance.
 - 02 **Focus marketing efforts on flagship products** and high-performing footwear categories.
 - 03 **Increase promotion of unisex footwear products** due to strong customer demand
 - 04 **Continue optimizing online and retail store channels** since both contributed significantly to sales performance.
 - 05 **Maintain product quality and customer satisfaction** to preserve strong brand ratings.
 - 06 **Conduct further analysis on monthly sales fluctuations** to identify seasonal sales opportunities.
-



Thank You



THANK YOU
